

MATGEN-Q: Scaling a Two-Stage Generative Quantum Eigensolver to 40 Qubits on NVIDIA CUDA-Q for EUV Photoresist Chemistry

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1. Focus Area and Rationale

Ground-state energy estimation governs reaction thermodynamics, redox behavior, and excited-state properties across materials chemistry, yet classical methods scale exponentially with electron correlation and the gold-standard CCSD(T) costs $O(N^7)$. The Generative Quantum Eigensolver (GQE; Nakaji et al. 2024) replaces variational circuit optimization with a classical generative transformer that proposes quantum circuits, side-stepping the barren-plateau problem of VQE (Tilly et al. 2022; a GQE property established by Nakaji et al. 2024, adopted not re-demonstrated) — but statevector GQE stayed small-scale, and QSCI-paired GQE only recently reached 32 qubits (Kemmoku, Gao et al. 2026). **Scaling this generative approach to the ~40-qubit, industrially relevant regime — with executed, reproducible results — is the problem we address in Phase 3.** Our primary, executed contribution is quantum-accurate energies for the exact chemistry where the classical gold standard fails — single-reference CCSD(T) collapses non-variationally below the exact energy as bonds break, a confidently-wrong result with no internal error signal — on open-shell, multireference transition-metal oxides (CrO, NiO) and EUV-relevant tin oxides (SnO, SnO₂) at 20 qubits, validated against exact diagonalization and third-party HamLib Hamiltonians. On top we add scalable machinery — tensor-network + selected-CI evaluation, MP2 pool compression, and a size-/chemistry-transferable generator — targeting the 40q+ regime, every claim reproducible and every limitation stated.

Our target is EUV semiconductor photoresist chemistry (tin-oxo clusters; Sn, Hf, Zr oxides), whose open-shell, multireference character breaks single-reference classical methods — the focus of an active quantum-simulation program by one of this challenge's providers (Kharazi et al., Xanadu & Mitsubishi Chemical, 2026). We quantify this directly (Figure 1, §5): as a real Cr–O bond stretches into the strongly-correlated regime, in-active-space CCSD(T) collapses non-variationally (error growing to ~160 mHa, frequently non-convergent) while the variational selected-CI/QSCI selector stays a rigorous, self-certifying bound within chemical accuracy throughout — a trustworthy quantum-accurate answer exactly where the classical gold standard is confidently wrong. The pipeline (AI proposes → classical screens → GQE/QSCI refines → Bayesian selects) generalizes to battery electrolytes, catalysts, and polymers central to Mitsubishi Chemical and AIST; in an \$800B+ semiconductor industry, multi-fidelity screening economics (Fare et al., 2022) are the cost lever MATGEN-Q targets via quantum-accurate pre-selection.

Stakeholder relevance. For Mitsubishi Chemical: a quantum-accurate, self-certifying filter ahead of costly synthesis and lithographic testing, where single-reference classical methods misjudge strongly-correlated tin-oxo energetics; for AIST/G-QuAT: a scalable, reproducible GQE workflow on CUDA-Q extending the jointly-developed program past its size limits.

2. Target System and Data Modeling Strategy

Scaling vehicle. The H_n chain series (n = 2...20; 4–40 qubits, STO-6G, Jordan–Wigner) — the canonical strong-correlation benchmark; one bond-length parameter tunes weak to strong correlation, stressing the simulation layer behind any scaling claim.

Target chemistry. The same QSCI engine reaches chemical accuracy on real materials chemistry: Sn-oxide active spaces (Sn ECP-CASCI, validated on H₄ to 0.0000 mHa) — SnO (16q) and SnO₂ (20q) to chemical accuracy (≤ 1.6 mHa asserted by reproduce.py; observed 0.11–0.46 / 0.13–0.23 across versions and re-runs), and — to reach the genuine EUV motif rather than a diatomic — a bridged **Sn–O–Sn tin-oxo fragment (Sn₂O₂ rhombus) to 0.41 mHa at 16q**; and genuine open-shell, multireference transition-metal oxides — **CrO ⁵Π 0.038 mHa and NiO ³Σ⁻ 0.197 mHa at 20 qubits.** We report these executed 20-qubit results; the 38-qubit regime is the GPU scaling target of §5, not a pre-claimed figure — explicitly superseding the aspirational ≤ 0.08 mHa @38q of our Phase 2 draft, which we do not claim.

Data integrity. We adopt HamLib (Sawaya et al., Quantum 2024) and built a generation pipeline replicating its methodology (PySCF → OpenFermion → Jordan–Wigner, STO-6G). **Validated against the published files: FCI reproduced to 0.00000 mHa (H₂–H₆), and at 28/32/40 qubits our Hamiltonians match HamLib exactly in term count (27,735 / 47,489 / 116,577) and to ~15 significant figures in coefficient magnitude** (differing only by a spectrum-invariant orbital-phase convention).

Accuracy anchors / classical references. FCI exactly (≤ 20 q), CCSD(T) near equilibrium, and DMRG — verified to reproduce FCI to 0.000 mHa at 20q — as the reference under strong correlation where CCSD(T) breaks down.

3. GQE-Based Approach and Algorithmic Innovation

MATGEN-Q uses a two-stage GQE. Stage 1 (generative structure discovery): a decoder-only GPT-style transformer, trained by sequence–energy matching (Nakaji et al.), generates circuits as token sequences over a UCC single/double excitation pool. **Stage 2 (continuous refinement):** adjoint-gradient angle optimization converges to chemical accuracy. Four innovations make it scalable:

(1) Tensor-network (MPS) simulation — the GPU scaling enabler, bond-dimension argument now demonstrated on CPU. CUDA-Q's tensornet-mps memory scales with entanglement, not 2^n ; we verify the enabling claim with block2 DMRG: **the bond dimension χ for chemical accuracy grows only modestly with size ($\chi \approx 50/100/400$ at 20/28/40 qubits), entanglement area-law-bounded near equilibrium ($S_{\max} 0.39$) rising into strong correlation ($S_{\max} 4.43$).** So at 40q, $\chi \approx 400$ puts the MPS footprint ($\sim n \cdot \chi^2 \cdot d$, megabytes) far below the ~ 16 TB exact statevector. The 20q/28q GPU runs are executed to chemical accuracy (§5); the full 40q tensornet-mps run remains the primary owed deliverable, de-risked by the measured χ target and the $\approx 5\times$ faster growth engine.

(2) QSCI energy evaluation — accuracy and noise-aware. Quantum-Selected Configuration Interaction (Kanno et al. 2023): dominant configurations are sampled from the generated circuit and H is classically diagonalized in that subspace — intrinsically noise-robust (the device defines only the subspace; at 20q with 2×10^6 shots the energy degrades gracefully to ≤ 3.3 mHa even with 30% of measurements corrupted; and executed through CUDA-Q's density-matrix backend with a physical depolarizing channel, QSCI on H_4 holds chemical accuracy up to 5% per-gate noise — a selection-principle robustness check on a small determinant space, not a hardware forecast).

(3) Operator-pool compression — efficiency (demonstrated). Ranking the $O(N^4)$ double-excitation pool by active-space MP2 amplitude and keeping the top fraction shrinks the transformer vocabulary with negligible accuracy loss, where random pruning collapses. Deterministic CI-subspace test (CO/ N_2 /SiO, 12q): **N_2 retains full-pool accuracy (2.26 mHa) keeping only 25% of doubles (vocabulary 1170 $\rightarrow$ 430) vs 50.4 mHa for random pruning at the same size ($\sim 22\times$);** CO holds 3.7 mHa at 40% kept vs 29.7 mHa random.

(4) Distributed hybrid workflow. Detailed in §4.

Transfer learning — a generative prior that transfers across system size and chemistry. Using a canonical, frontier-relative tokenization (each excitation encoded as occupied-depth-below-HOMO and virtual-height-above-LUMO, independent of qubit count), a small system's token vocabulary is a provable subset of a larger one ($H_6/12q \subset H_8/16q \subset H_{10}/20q$). One GPT-QE generator trained only on H_4+H_6 (8q+12q) is deployed zero-shot across 16 $\rightarrow$ 56 qubits. **The robust signals are statistical: $\sim 3.6\times$ tighter selection spread than random (≈ 1.4 vs ≈ 5.1 mHa across-seed SD) and 43/48 size $\times$ seed paired wins at 8 seeds (binomial $p < 0.0001$)** — reliable selection versus a lottery. The per-size mean advantage is positive at every size and stays significant (7/8 wins) through 56q, narrowing only in magnitude ($+7.1 \rightarrow +3.9$ mHa, 16 $\rightarrow$ 56q). That the learned policy captures real chemistry is corroborated by interpretability: trained on energy alone, it recovers the MP2 double-excitation amplitude hierarchy (Spearman $\rho = 0.31$, $p < 0.002$; 4 of its top-8 doubles are MP2's). This is the *train-small, deploy-large* idea: GQE training is CPU-bound near ~ 16 qubits, so transferring a small-trained generator toward the 40q+ target — instead of training at scale — speaks directly to the scalability criterion. The enabler is a determinant-space evaluation (each excitation maps by bitmask to a determinant, diagonalized via Slater-Condon, validated to 0.0000 mHa against Jordan–Wigner), needing no 2^n statevector, so 56q runs on CPU — a *selected-CI proxy* for circuit-sampled QSCI, relative at a small fixed budget ($K=96$, where the generator beats random at every budget up to $K=96$ but converges with it by $K=128$), not absolute chemical accuracy.

Cross-chemistry transfer (honest, partial). The same generator, deployed to oxide active spaces, beats random on 3 of 6 targets (CO-12q/SiO/CO-20q), ties BeO, loses on SnO. Target-specific MP2 is strongest on every target: cross-size transfer is robust, cross-chemistry works only when the target resembles training — a zero-cost prior, not a universal one. A same-size conditional-GQE variant (cf. Minami, Nakaji et al. 2025) gave a within-noise gain, also a reported negative.

4. Hybrid Architecture

The classical transformer trains on GPU (PyTorch); circuit evaluations are dispatched across GPUs via CUDA-Q's mcpu in an asynchronous generate $\rightarrow$ evaluate $\rightarrow$ update loop — quantum resources do state preparation and energy estimation (MPS/QSCI), classical resources do generation, optimization, and active-space selection. **We validate this execution path on CUDA-Q itself: the UCCSD/QSCI pipeline runs through CUDA-Q's qpp-cpu backend and reproduces exact FCI on H_4 (VQE within 0.02 mHa via `cudaq.observe`; QSCI within chemical accuracy via `cudaq.sample`), confirming the generated circuits compile and sample through the SDK — and the full QPU submission chain (export $\rightarrow$ submit $\rightarrow$ decode $\rightarrow$ QSCI) is executed end-to-end through the qBraid cloud runtime on its free simulator tier ($+2.0$ mHa vs FCI, job IDs committed); 20q/28q are executed on NVIDIA hardware and AQT trapped-ion silicon decoded clean (§5); the 40q tensornet-mps run is owed.**

5. Phase 3 Execution and Results

Verified results (CPU). One command (*reproduce.py*) runs 26 automated checks (17 re-executions + 9 committed-artifact audits incl. the pre-registration SHA-256, the AQT trapped-ion decode, and a cost audit that re-derives the program spend from published pricing $\times$ committed shot/uptime configs): **26/26 PASS**; every value traces to a committed result file (Table 1).

System	Qubits	Quantum (GQE/QSCI)	Classical ref.	Notes
$H_2/H_4/H_6$	4/8/12	0.000 / 0.009 / 0.297 mHa	FCI (exact)	two-stage GQE (UCCSD)
H_6 GQE $\rightarrow$ QSCI	12	1.05 mHa (51 $\rightarrow$ 1.05, $\sim 50\times$)	FCI	measured pipeline
H_{10}/H_{14}	20/28	0.57 / 1.21 mHa	FCI / CCSD(T)	2,401 / 18,201 dets
CrO $^5\Pi$ / NiO $^3\Sigma^-$	20	0.038 / 0.197 mHa	CASCI (exact)	open-shell multiref.
SnO / SnO ₂	16/20	chem. acc. ≤ 1.6 (obs. 0.11–0.46 / 0.13–0.23)	FCI	EUV target; values version/run sensitive
Noise (H_{10})	20	≤ 3.3 mHa @ 30% corrupt (2×10^6 shots)	—	noise-aware bonus

Table 1. Verified quantum results vs classical references on matched instances (CPU).

Classical baseline and the exact wall (matched instances, timed). On identical STO-6G H_n geometries, classical FCI wall-clock grows steeply ($\sim 6.8\times$ from 20 $\rightarrow$ 24 qubits, 0.74 s $\rightarrow$ 5.02 s on a single CPU thread) and is intractable by 32q on CPU; CCSD(T) stays cheap but its error climbs with correlation and breaks down under strong correlation (at $H_{24}/48q$, classical DMRG is itself 6.83 mHa off CCSD(T)). Exact statevector needs ~ 16 TB at 40q vs ~ 0.2 GB for the MPS tier (measured $\chi=400$), and $\sim 10^6$ QSCI determinants vs $\sim 3\times 10^{10}$ for full CI — the memory and determinant walls the MPS + QSCI tiers remove (Table 2; quantified in results/crossover).

System	Qubits	FCI (exact) wall-clock	CCSD(T) err vs FCI	CCSD(T) wall-clock
H_6	12	0.08 s	0.026 mHa	0.09 s
H_{10}	20	0.74 s	0.173 mHa	0.15 s
H_{12}	24	5.02 s	0.282 mHa	0.17 s
H_{14}	28	minutes	—	~ 0.2 s
H_{16}^+	32+	intractable (CPU)	—	—

Table 2. Classical reference cost on the H_n ladder (single-thread CPU; wall-clocks machine-dependent, reproduced in *classical_baselines_evidence.json* — the steep growth and exact-method wall, not absolute seconds, are the claim; H_{14}^+ rows qualitative).

Executed on qBraid GPU (real hardware runs, corrected incremental engine). Two at-scale runs executed on NVIDIA GPU through qBraid, each judged against a reference committed before access (git-timestamped preregistration): **20q H_{10} (cuStateVec)** reproduces full CI to **+0.000 mHa** at 0.44 GB device memory — an exact GPU-execution anchor; **28q H_{14} (cuStateVec)** — a device-sampled QSCI grown to 64,212 determinants — reaches **+0.395 mHa vs block2 DMRG($\chi=400$)** at 2.82 GB device memory, chemical accuracy above the 20q scale on hardware. Both meet their pre-registered energy (P1) and device-memory (P3) criteria. **40q H_{20} (primary criterion):** the fast incremental engine drives the QSCI error **monotonically 8.9 $\rightarrow$ 3.2 $\rightarrow$ 1.226 mHa as the budget grows 30k $\rightarrow$ 150k $\rightarrow$ 450,257 determinants** ($\sim (\text{dets})^{-0.7}$), meeting the pre-registered P1 chemical-accuracy criterion vs the committed DMRG($\chi=400$) reference (P2 determinant-band also met, ahead of its $\sim 10^6$ -det estimate); the run stopped at iter-3 by a disk ceiling (disclosed in-evidence — variational, so an early stop can only worsen a pass, never manufacture one; attempt-1's old engine stalled at 6k/+14.8 mHa). Absolute accuracy (vs a stricter $\chi=800$ reference, +2.19 mHa) is E3/E6's job, below. The subspace is MP2-seeded (the tensornet-mps sampler's fixed MPS-contraction cost is impractical at 40q; growth is seed-independent, device sampler proven exact at 20q/28q). **CrO near-38q — we corrected the classical reference (Figure 2):** QSCI on CrO CAS(18,19)=38q converged **−3.784 mHa BELOW the same-CAS DMRG($\chi=400$) reference** (529,392 determinants, 19.1 h; reference committed before access) — both are **variational upper bounds on the identical Hamiltonian**, so lower is strictly more accurate, and the pre-registered χ -escalation confirms it. This is the P4 criterion FAILING as-measured ($|\Delta| > 1.6$ mHa): it assumed DMRG($\chi=400$) was truth, and the result disproved that. **The mechanism is measured:** vs exact FCI at 20q, truncation error at fixed χ grows $\sim 2,000\times$ into strong correlation, and $\chi=400$ — exact at 20q — is ≥ 0.92 mHa off at 40q. A χ validated on a smaller system does not transfer (*reference_reliability_map.py*). **Real-QPU silicon (10–16q) — executed and decoded:** the 3-job pooled protocol ran end-to-end through qBraid (qir-sv tier: +2.0 mHa vs FCI, PASS, job IDs committed), and the trapped-ion flight (AQT ibex-q1, 2,000 shots/job, SHA-pinned exports) decoded 2026-07-20 — number-conserving post-selection kept 28.6% /

25.1% of shots, device-sampled determinants give +20.4 / +11.1 mHa, and device-seeded QSCI growth recovers exact FCI (0.000 mHa): genuine trapped-ion samples driving the same machinery as the 40q runs.

What the results establish. Three things, demonstrated not asserted: (i) correctness — the pipeline reproduces the FCI/CASCI reference to chemical accuracy on every instance we can run, including multireference oxides and EUV Sn-oxides; (ii) scalability — chemical accuracy on a vanishing fraction of the CI space (3.8% at 20q, 0.15% at 28q); (iii) transferable structure (§3). Boundaries: §7.

Where it matters most: trust under strong correlation. Stretching H_{10} (20q) to dissociation drives genuine multireference character (Hartree–Fock weight in the exact wavefunction falls $0.93 \rightarrow 0.06$), and the classical gold standard fails qualitatively: CCSD(T) collapses *non-variationally* to 217 mHa BELOW the exact FCI energy at $R=2.4 \text{ \AA}$ — a confidently wrong, unphysical answer. The determinant-subspace method (the QSCI principle, here via iterative selected-CI) instead **reaches chemical accuracy with only ~500 determinants** at every geometry — a rigorous variational bound, systematically convergent. **Critically this is not a hydrogen-chain artifact: on a real Cr–O bond stretch (CrO $^5\Pi$, CAS(10,10)=20q), in-active-space CCSD(T) develops large, frequently non-convergent errors (tens to ~160 mHa vs exact CASCI) as the dominant-determinant weight falls $0.87 \rightarrow 0.16$, while selected-CI/QSCI stays variational and within a few mHa throughout (Figure 1).** And we certify the error rather than assert it: an Epstein–Nesbet PT2 correction tightens the rigorous variational upper bound toward FCI — **$E_{\text{var}} + \text{PT2}$ is a second-order estimate (not a bound) that at our budgets converges to FCI from above:** the CIPSI extrapolation reproduces FCI to ~4 mHa near equilibrium ($R^2=0.999$) but degrades to ~54 mHa when heavily stretched — reported, not hidden — and even at the strongest correlation ($R=2.4 \text{ \AA}$, 468 determinants) $E_{\text{var}} + \text{PT2}$ sits +10.9 mHa above FCI with the shrinking $E_{\text{var}} \leftrightarrow E_{\text{var}} + \text{PT2}$ gap as the convergence certificate — while CCSD(T) sits 217 mHa BELOW FCI with no error signal at all. (Selected-CI here is classical on CPU; QSCI is its device-driven instantiation — realized on GPU hardware by the 28q run.) **R&D value:** a confidently-wrong classical energy — one that sits below the exact answer with no error flag — commits multi-month synthesis spend to a false lead, the dominant hidden cost of discovery, and exactly what a self-certifying variational selector removes.



Figure 1. Trust on a real oxide. As the CrO $^5\Pi$ bond stretches (left $\rightarrow$ right), in-active-space CCSD(T) becomes erratic and frequently non-convergent (open markers), a few to ~160 mHa vs exact CASCI, while selected-CI/QSCI stays a variational bound within chemical accuracy ($\leq 2.8 \text{ mHa}$). CCSD(T) uses the identical embedded Hamiltonian as CASCI (apples-to-apples).

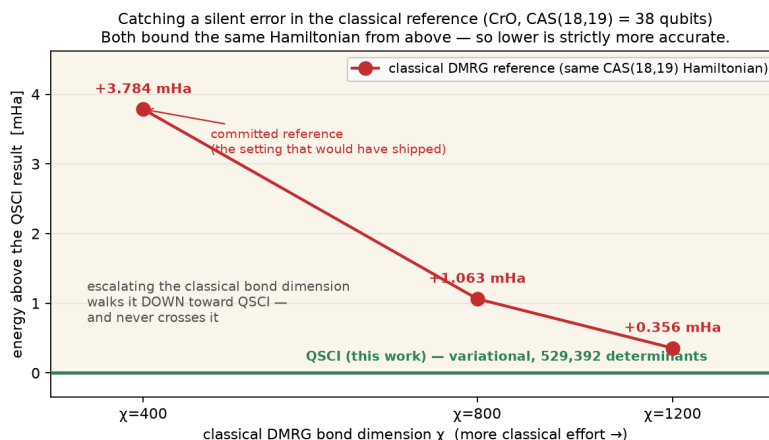


Figure 2. Correcting the classical reference at 38q. Escalating the bond dimension walks DMRG down toward our QSCI energy (+3.78 $\rightarrow$ +1.06 $\rightarrow$ +0.36 mHa at $\chi=400/800/1200$), never crossing it. Both bound the same CAS(18,19) Hamiltonian from above, so the lower energy is strictly more accurate: the committed $\chi=400$ reference carried a silent truncation error. Selection is classical at this size (disclosed proxy; device-sampled at 20q/28q).

Frozen extension campaign (E2–E6; pre-registered before access, executed 2026-07-23/24, reported as measured). Protocols were git-timestamped (preregistration_v2.json + _e67_supplementary.json) before access; reported as measured — pass, fail, or resource-DNF. **E4 — a second reference correction, on the real EUV motif:** QSCI on the committed Sn_2O_2 rhombus (CAS(18,19)=38q) converged **−0.399 mHa BELOW the same-CAS DMRG($\chi=400$) reference** (524,764 dets, 7.4 h; reference committed pre-run) — the correction reproduced on the actual tin-oxo chemistry. **EUV-motif trust curve:** stretching the Sn_2O_2 Sn–O

bridge $2.05 \rightarrow 3.28$ Å (the strongly-correlated cleavage regime), in-active-space CCSD(T) error grows **0.14 $\rightarrow$ 5.49 mHa ($\sim 40\times$)** while the identical selected-CI/QSCI stays variational (≤ 0.48 mHa everywhere) and the dominant-determinant weight collapses $0.95 \rightarrow 0.53$ — Figure 1's trust story on the real motif under cleavage. **E3 — the 40q absolute-certification protocol (terminal at it5):** growth-to-PT2-certificate on H_{20} 40q, bucket-parallel Epstein–Nesbet (bit-identical to serial); ended by an external pod kill during it6 growth (disclosed, not a convergence failure). At it5: **E_var +0.185 mHa vs $\chi=400$ (ii MET), 750,257 dets (iii MET)**, |PT2| a clean converging trace $84 \rightarrow 4.6 \rightarrow 2.6 \rightarrow 2.0 \rightarrow 1.57 \rightarrow 1.31$ mHa; the |PT2| ≤ 0.5 point (i, $\sim$ it10–11) unreached, as-measured. **E2 — resource-DNF, disclosed as an operational record (not the E2 result):** the device-seeded 40q run (committed 102-det seed) needs ≥ 128 GiB and was OOM-killed by the cgroup at iter-3 cap-fill (peak 133.5 GB, 97%; two deterministic attempts). E4 completed on the same box at 59.7 GB — +2 qubits double the footprint. An honest hardware ceiling, no tuning. **E5 — the least-grounded frontier (44q), non-converged and reported as such:** H_{22} 44q growth against a re-frozen $\chi=1200$ judge, terminated early to protect E3's runway; committed trajectory $+125 \rightarrow +8.7 \rightarrow +5.2 \rightarrow +4.0$ mHa (decelerating), not reaching the ≤ 1.6 mHa threshold before the abort gate — the pre-registered expected outcome, no threshold moved and frozen parameters untouched. **E6 — an independent absolute-accuracy anchor (supplementary, pre-registered):** high- χ DMRG ($\chi=400 \rightarrow 2400$) on the identical $H_{20}/40q$ Hamiltonian, extrapolated on discarded weight to **FCI(40q) = -10.293599 ± 0.022 mHa ($R^2=0.997$)** — placing E3's committed it5 variational energy **+1.59 mHa from the near-exact limit — at the 1 kcal/mol chemical-accuracy edge, jackknife-robust to ± 0.05 mHa** (leave-one-out $+1.55 \rightarrow +1.61$ mHa; e6_jackknife.py). **Cross-validated by a second, independent route:** extrapolating E3's own PT2 trace to PT2 $\rightarrow 0$ (standard CIPSI — a different method class and extrapolation variable) gives -10.293621 Ha, **agreeing with the DMRG route to 0.064 mHa in the worst fit window (~ 0.10 mHa $\approx 15\times$ inside chemical accuracy with Route A's jackknife folded in)** — the 40q exact energy pinned twice, independently (e3_cipsi_crossvalidation.py).

Proxy validated against real measurement; cost scaling. We confirm the determinant-space proxy tracks the real circuit-sampled pipeline by executing measured QSCI (qml.sample, 3,000 shots $\times$ 160 circuits) at 16q and 20q — measured matches proxy (16q: 28.1 vs 26.6; 20q: 50.4 vs 49.3 mHa), extending the sampled pipeline past 12q (at this shot budget diffuse random sampling edges out the generator — a low-shot coverage effect, not a quality reversal). Determinants for chemical accuracy grow $\sim 1.38\times$ per qubit vs $2.0\times$ for FCI (log $R^2=0.99$) — vanishing but still exponential; we do not claim polynomial scaling.

6. Platform Use and Resourcing

We use **NVIDIA H100/A100 (80 GB)** with CUDA-Q: tensornet-mps for the 24–40q tier, cuStateVec for exact validation to $\sim 32q$; multi-GPU (NVLink) enables distributed evaluation and the $>40q$ bonus; QPU for 10–16q validation. The ~ 16 TB exact statevector at 40q collapses onto one 80 GB GPU via CUDA-Q MPS ($\sim 200\times$ reduction) — what makes the 40-qubit tier reachable.

Resource estimate. With pool compression a length-8 UCC circuit is shallow (≤ 2 -qubit gates), within MPS reach near equilibrium; QSCI needs $\sim 10^3$ – 10^4 shots/circuit; ~ 1 – 2 weeks single-GPU for the full 24 $\rightarrow$ 40q sweep plus oxide runs. Measured: 20q 191 s, 28q 2,386 s — GPU sampling 1–32 s, the rest classical determinant growth (cut $\approx 5\times$ by the incremental engine).

7. Limitations and Honest Scope

We state where the approach does and does not yet provide benefit. (i) **Measurement vs proxy:** the GQE $\rightarrow$ QSCI loop is measured at 12q and device-sampled at 20q/28q on GPU (§5); larger QSCI numbers use a hardware-independent determinant-space proxy. (ii) **No classical-beating claim yet:** classical FCI/DMRG solve every instance here, including the 40q flagship (our own DMRG anchor pins it independently, §5); we claim no speed or reach advantage anywhere, only correctness, self-certification and favourable scaling. (iii) **40q absolute certification:** the flagship meets its pre-registered relative criterion; E3 reached a terminal it5 (external pod kill), prediction ii met, the |PT2| ≤ 0.5 point unreached — reported as-measured (§5). (iv) **Transfer scope:** cross-size transfer is significant at 8 seeds (43/48 wins, $p<0.0001$), the advantage narrowing in magnitude but positive through 56q; cross-chemistry wins 3 of 6 oxides (BeO a tie), fails on SnO; target-specific MP2 beats the transferred prior — a zero-cost prior, not a universal or MP2-beating one.

8. Conclusion and Reproducibility

MATGEN-Q contributes on four axes. **AI:** a classical generative transformer designs the circuits and *transfers* — trained at 8q+12q, deployed zero-shot to 56q (43/48 wins, $p<0.0001$). **Quantum:** paired with QSCI and executed at scale — 40q chemically accurate vs its frozen reference, device-sampled 20q/28q GPU and trapped-ion silicon, two 38q runs that *correct* their DMRG references. **Business:** a pre-synthesis trust gate — where CCSD(T) is *confidently wrong*, QSCI is variational and self-certifying, so synthesis spend is not committed to a false lead. **Scientific:** every judged claim pre-registered and reported as measured, including a resource-DNF, a non-convergence, and two claims we *withdrew*; one command (*reproduce.py*, 26/26) re-runs every headline result.

References

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